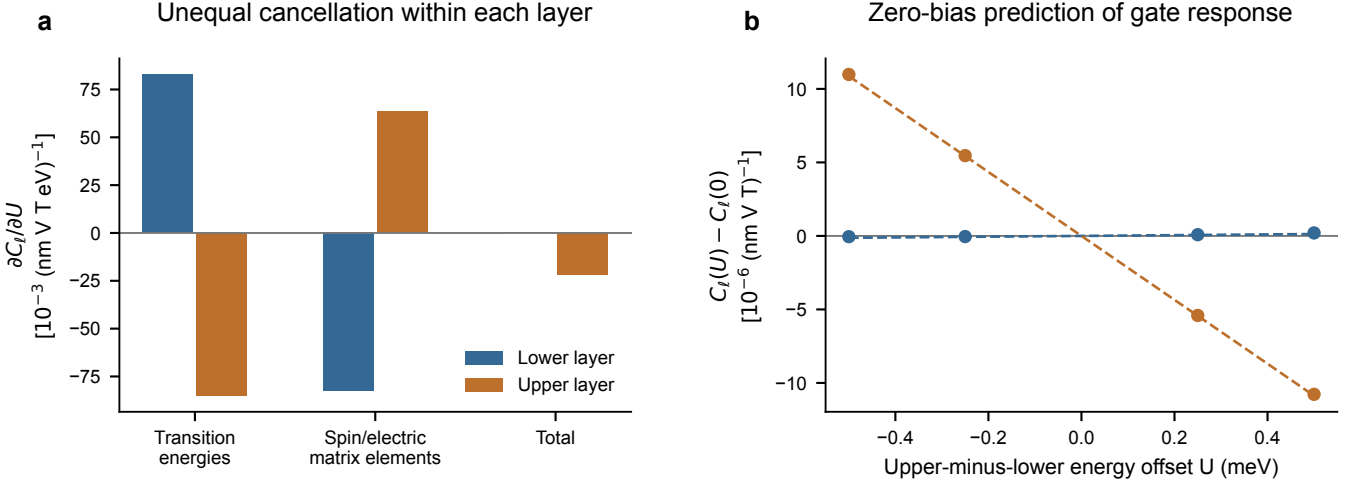Bilayer KM: $s_{\parallel,\ell} = C_\ell E_\parallel B_z$